

Lab Sheet # L9

Instructions:

- Submit *.sql file, named as Lab09_6XXXXXX.sql to LAs or instructors prior to My Courses.
- If students cannot finish within the class time, students can submit until the beginning of the next class.

Self-study:

1. View: <https://www.mysqltutorial.org/mysql-views/>
2. Index: <https://www.mysqltutorial.org/mysql-index/>

Today's Task

0. Database Setup:
 - a. Import the database by executing the **PremierProducts.sql** and **Data_PremierProducts.sql** files.

Easy Task: Basic VIEW / basic INDEX

1. Create a **VIEW** named **view_customer_basic** that displays CustomerNum, CustomerName, City, State from table customer.

20 • `SELECT * from view_customer_basic;`

	CustomerNum	CustomerName	City	State
▶	148	Al's Appliance and Sport	Fillmore	FL
	282	Brookings Direct	Grove	FL
	356	Ferguson's	Northfield	FL
	408	The Everything Shop	Crystal	FL

2. Create a **VIEW** named **view_part_price** showing PartNum, Description, Price.

32 • `SELECT * from view_part_price;`

	PartNum	Description	Price
▶	AT94	Iron	24.95
	BV06	Home Gym	794.95
	CD52	Microwave Oven	165.00
	DL71	Cordless Drill	129.95

3. Create a **VIEW** named **view_order_basic** showing

OrderNum, OrderDate, CustomerNum.

44 • `SELECT * from view_order_basic;`

	OrderNum	OrderDate	CustomerNum
▶	21608	2010-10-20 00:00:00	148
	21610	2010-10-20 00:00:00	356
	21613	2010-10-21 00:00:00	408
	21614	2010-10-21 00:00:00	282

4. Create a **VIEW** named **view_rep_contact** showing

RepNum, FirstName, LastName, City.

57 • `SELECT * from view_rep_contact;`

	RepNum	FirstName	LastName	City
▶	20	Valerie	Kaiser	Grove
	35	Richard	Hull	Sheldon
	65	Juan	Perez	Fillmore

5. Write SQL to display the SQL definition of **view_customer_basic**.

View	Create View	character_set_client	collation_connection
▶ view_customer_basic	CREATE ALGORITHM=UN...	utf8mb4	utf8mb4_0900_ai_ci

6. Write SQL to remove the view **view_part_price**.

70 • `SHOW FULL TABLES WHERE Table_Type = 'VIEW';`

Tables_in_premierproducts	Table_type
▶ view_customer_basic	VIEW
view_order_basic	VIEW
view_rep_contact	VIEW

7. Create an **INDEX** named **idx_customer_name** on column **CustomerName** in table **customer**.

81 • `SHOW INDEX FROM customer;`

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type
customer	0	PRIMARY	1	CustomerNum	A	2				BTREE
customer	1	RepNum	1	RepNum	A	2			YES	BTREE
customer	1	idx_customer_name	1	CustomerName	A	10			YES	BTREE

8. Create an **INDEX** named `idx_order_date` on column `OrderDate` in table `orders`.

91 • `SHOW INDEX FROM orders;`

Table	Non_unique	Key_name	Seq_in_index	Column_name
orders	0	PRIMARY	1	OrderNum
orders	1	CustomerNum	1	CustomerName
orders	1	idx_order_date	1	OrderDate

9. Create an **INDEX** on column `PartNum` in table `orderline`.

99 • `SHOW INDEX FROM orderline;`

Table	Non_unique	Key_name	Seq_in_index	Column_name
orderline	1	OrderNum	1	OrderNum
orderline	1	idx_orderline_part	1	PartNum

10. Show all indexes in table `part`.

Table	Non_unique	Key_name	Seq_in_index	Column_name
part	0	PRIMARY	1	PartNum

Medium Task: VIEW with JOIN / GROUP BY

11. Create a **VIEW** named `view_order_customer` that shows

`OrderNum`, `OrderDate`, `CustomerName`.

Tables: `orders`, `customer`

121 • `SELECT * FROM view_order_customer;`

OrderNum	OrderDate	CustomerName
21608	2010-10-20 00:00:00	Al's Appliance and Sp
21610	2010-10-20 00:00:00	Ferguson's
21613	2010-10-21 00:00:00	The Everything Shop
21614	2010-10-21 00:00:00	Brookings Direct

12. Create a **VIEW** named **view_order_detail** showing

OrderNum, PartNum, Description, NumOrdered

Tables: orderline, part

```
137 • SELECT * FROM view_order_detail;
```

OrderNum	PartNum	Description	NumOrdered
21608	AT94	Iron	11
21610	DR93	Gas Range	1
21610	DW11	Washer	1
21613	KL62	Dryer	4

13. Create a **VIEW** named **view_customer_rep** that shows

CustomerName, Rep FirstName, Rep LastName

Tables: customer, rep

```
152 • SELECT * FROM view_customer_rep;
```

CustomerName	RepFirstName	RepLastName
Al's Appliance and Sport	Valerie	Kaiser
Kline's	Valerie	Kaiser
All Season	Valerie	Kaiser
Brookings Direct	Richard	Hull

14. Create a **VIEW** named **view_order_total** showing

OrderNum, TotalAmount

Where: $TotalAmount = SUM(NumOrdered * QuotedPrice)$

Tables: orderline

Grouped by: OrderNum.

```
166 • SELECT * FROM view_order_total;
```

OrderNum	TotalAmount
21608	241.45
21610	894.99
21613	1319.80
21614	1190.00

15. Create a **VIEW** named `view_customer_total_orders` showing

CustomerName, TotalOrders

183 • `SELECT * FROM view_customer_total_orders;`

CustomerName	TotalOrders
Al's Appliance and Sport	2
Brookings Direct	1
Ferguson's	1
The Everything Shop	1

Hard Task: Design view + indexing strategy

16. Create a **VIEW** named `view_customer_purchase`

Showing: CustomerName, OrderNum, PartNum, NumOrdered, QuotedPrice

Tables required: customer, orders, orderline

216 • `SELECT * FROM view_customer_purchase;`

CustomerName	OrderNum	PartNum
Al's Appliance and Sport	21608	AT94
Al's Appliance and Sport	21619	DR93
Brookings Direct	21614	KT03
Ferguson's	21610	DR93

17. Create a **VIEW** named `view_top_customer_sales`

Showing: CustomerName, TotalSales

Where: $\text{TotalSales} = \text{SUM}(\text{NumOrdered} * \text{QuotedPrice})$

Group by: customer

237 • `SELECT * FROM view_top_customer_sales;`

CustomerName	TotalSales
Johnson's Department Store	4769.90
The Everything Shop	1319.80
Brookings Direct	1190.00
Ferguson's	894.99

18. A query frequently searches orders by `CustomerNum`, `OrderDate`

Write SQL to create an index to optimize this query. (No Expected output)